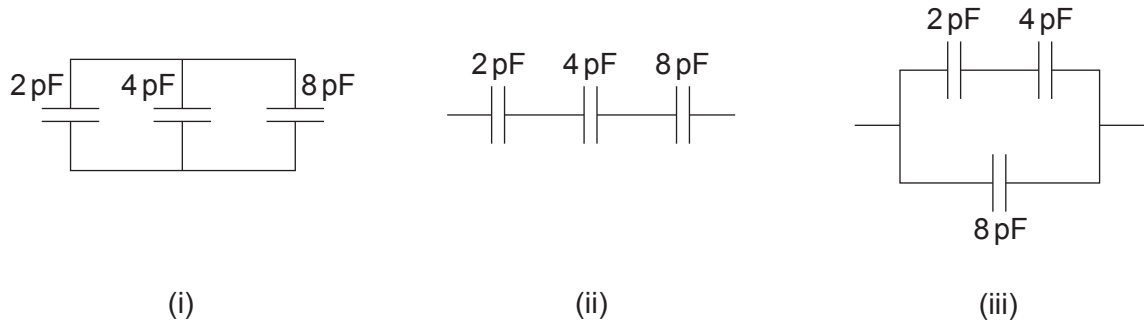


SECTION A

Answer **all** questions.

1. (a) Calculate the total capacitance of each of the following capacitor combinations. [6]



(i) .....

.....

.....

(ii) .....

.....

.....

(iii) .....

.....

.....



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only

- (b) Michael, a laboratory technician, must make a  $2.7\text{ }\mu\text{F}$  capacitor using only two square sheets of aluminium of side  $42.0\text{ cm}$ . Determine whether or not it is realistic for Michael to produce a capacitance of  $2.7\text{ }\mu\text{F}$  using these two sheets of aluminium and no dielectric. [4]

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- (c) State what happens to the capacitance of a parallel plate capacitor when a dielectric is placed between the plates. [1]

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